

Three months of continuous monitoring of atmospheric water vapor with a network of Global Positioning System receivers

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Abstract. Three months of continuous data from the Global Positioning System (GPS) using 20 sites in Sweden and 5 sites in Finland have been used to estimate the integrated amount of atmospheric water vapor. The quality of the data has been assessed by comparisons with a microwave radiometer (water vapor radiometer (WVR)) at the Onsala Space Observatory and with data from four different radiosonde stations. We found the agreement in integrated water vapor (IWV) between the GPS estimates and the radiometer data to be 1–2 kg/m² in terms of daily root-mean-square (rms) differences. A major part of these rms differences were caused by a bias between the data sets. This bias (WVR–GPS) varied from day to day between –1.0 and +2.5 kg/m² with a mean value of +1.3 kg/m². Comparisons with radiosonde data showed rms differences around or slightly above 2 kg/m² for each station using the entire 3 month data set. Also here the GPS estimates were, on the average, below the radiosonde results. We show that the radomes used to protect the GPS antennas are likely to cause a large part of the observed bias. Spatial structure functions were calculated by using the GPS and the radiosonde data. An overall consistency between the GPS-based and the radiosonde-based structure functions indicates that the spatial correlations between the GPS estimates are not affected by the estimation process used in the GPS data analysis.

1. Introduction

A method to estimate integrated water vapor (IWV) in the atmosphere using data from the Global Positioning System (GPS) has been described by, for example, *Bevis et al.* [1992]. Work by *Kuo et al.* [1993, 1996] used simulated water vapor data to investigate the possibilities of retrieving the vertical moisture resolution. They also studied the effect of IWV assimilation on short-range precipitation forecasts. From these investigations they found that assimilation of IWV data could effectively recover the vertical structure of the water vapor. Their results also indicated that when temperature and wind observations were available, the inclusion of IWV and surface humidity data clearly improved the precipitation forecast. Since the water vapor is an effective greenhouse gas, this type of GPS data also have a potential value for climate studies.

The accuracy of the technique to estimate the IWV with GPS has been assessed in several experiments over time periods spanning up to 40 days using different kinds of networks with up to 11 GPS receivers [e.g.,

Dixon and Kornreich, 1990; *Rocken et al.*, 1993, 1995; *Duan et al.*, 1996; *Coster et al.*, 1996]. In the study presented here, we have estimated the IWV using a relatively dense network (shortest baseline $\approx$ 100 km) with data from 20 of the continuously operating GPS stations in Sweden and 5 sites in Finland, belonging to the Finnish permanent GPS network. The latter data were obtained from the Finnish Geodetic Institute (FGI). The study has been concentrated in time to the period August to October 1995. We evaluate the quality of the GPS data by making comparisons with microwave radiometer data from the GPS site at the Onsala Space Observatory as well as radiosonde data from sites close to the GPS antennas. The data presented in this paper will also be used in the evaluation of forecast and climate models for the catchment area of the Baltic Sea within the BALTEX experiment [*Isemer*, 1995].

The principal observable in GPS is the signal travel time between the satellite and the receiving antenna. Both the code and the phase measurements can be used to determine the signal travel time, but for geodesy purposes only the phase measurements are precise enough. The phase of the radio waves transmitted from the satellites are delayed in the atmosphere. This delay is caused by the ionosphere, and the neutral atmosphere. The

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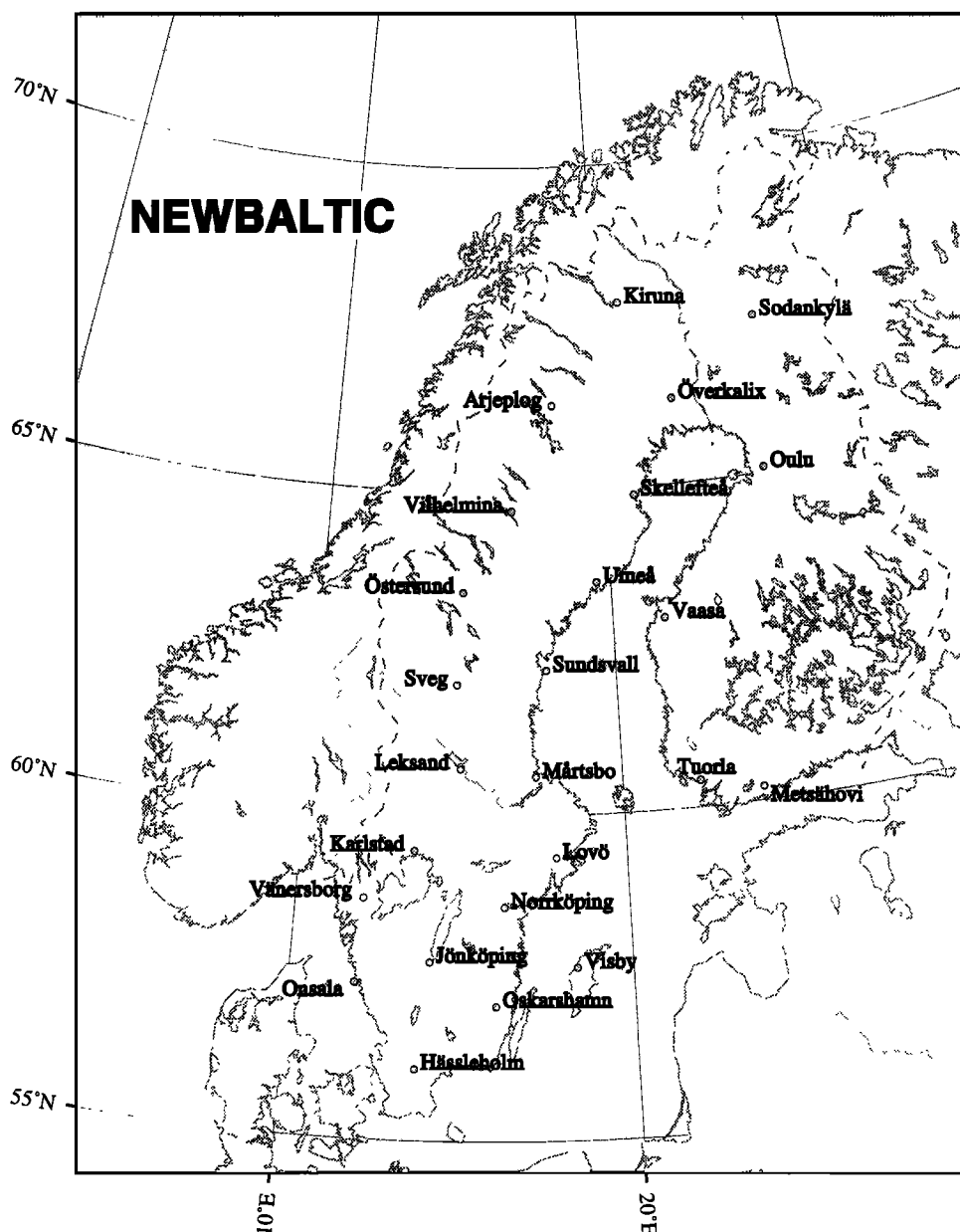


Figure 1. Swedish GPS network (SWEPOS) together with five Finnish GPS sites.

ionospheric delay is dispersive, and the effect can be well determined since the satellites are transmitting at two different frequencies [Seeber, 1993]. The neutral delay is usually divided in two parts, the hydrostatic delay due to the induced dipole moment and the wet delay mainly due to the permanent dipole moment of water vapor [Davis et al., 1985; Elgered, 1993]. Using a filtering technique it is possible to separate the total neutral delay, in the zenith direction, from the other estimated parameters such as position and clock errors. The hydrostatic delay in the zenith direction can then be determined by using pressure data according to Saastamoinen [1972]. With this quantity removed from the total delay, we have the zenith wet delay. The zenith wet delay is strongly correlated with the integrated wa-

ter vapor (IWV), and different methods exist describing this relation [e.g., Bevis et al., 1992, 1994].

The GPS technique is today commonly used for surveying and very accurate position determinations. In geodesy these measurements are being used for, for example, determinations of regional deformation due to plate tectonics [e.g., Blewitt et al., 1993; Bock et al., 1993; Dragert and Hyndman, 1995; Tsuji et al., 1995] and postglacial rebound [e.g., BIFROST project, 1996]. For background information about the GPS technique and the application of atmospheric remote sensing, see, for example, Seeber [1993], Hofmann-Wellenhof et al. [1994], and Davis et al. [1996].

In sections 2 and 3 we will give a brief presentation of the different techniques used. Thereafter, we will

Table 1. Sites in the GPS Network and Their Availability

GPS Site		Site Coordinates			Pillar Type ^a /Height m	Radome Type ^b	Number of Days With Data
Acronym	Full Name	Longitude °E	Latitude °N	Height m			
KIRU	Kiruna	21.060	67.877	469	C/3	W	85
SODA	Sodankylä	26.389	67.421	279	M/3	D	82
ARJE	Arjeplog	18.125	66.318	459	C/3	W	89
OVER	Över Kalix	22.770	66.310	200	C/3	W	92
OULU	Oulu	25.893	65.086	71	M/3	A	31
SKEL	Skellefteå	21.050	64.880	60	C/3	W	90
VILH	Vilhelmina	16.559	64.697	421	C/3	W	86
UMEA	Umeå	19.509	63.578	33	C/3	W	91
OSTE	Östersund	14.858	63.442	460	C/3	W	92
VAAS	Vaasa	21.771	62.961	40	M/3	N	35
SUND	Sundsvall	17.659	62.232	6	C/3	W	92
SVEG	Sveg	14.700	62.017	459	C/3	W	89
LEKS	Leksand	14.877	60.722	448	C/3	W	91
MART	Mårtsbo	17.258	60.595	50	C/3	W	91
TUOR	Tuorla	22.443	60.416	41	M/3	N	86
METS	Metsähovi	24.395	60.217	76	M/25	N	88
KARL	Karlstad	13.505	59.444	82	C/3	W	89
LOVO	Lovö	17.830	59.340	58	C/3	W	91
VANE	Vänernborg	12.070	58.690	136	C/3	W	90
NORR	Norrköping	16.250	58.590	14	C/3	W	88
JONK	Jönköping	14.059	57.745	228	C/1	W	85
VISB	Visby	18.367	57.653	55	C/3	W	90
ONSA	Onsala	11.925	57.395	10	C/1	D	91
OSKA	Oskarshamn	16.000	57.060	121	C/3	W	92
HASS	Hässleholm	13.718	56.092	78	C/3	W	89

^aM, metal; C, concrete.^bA, Ashtech; D, Delft; W, White; N, None.

present results and comparisons. In section 5 we discuss some of the possible errors that can affect our estimates and conceivable solutions. Finally, conclusions and future work are dealt with in section 6.

2. GPS Analysis

Figure 1 shows the geographical locations of the 25 GPS sites. All the stations with their latitude, longitude, and the number of days with GPS data are displayed in Table 1. Two of the Finnish sites have significantly less data; Vaasa and Oulu were taken into operation in late September 1995.

We processed the GPS data from this network with the GIPSY/OASIS software [Webb and Zumbege, 1993]. The GIPSY software uses a Kalman filter technique to estimate the unknown parameters. In this study we incorporated precise orbits distributed by the International GPS Service for Geodynamics (IGS) [Beutler et al., 1996; Noll and Gurtner, 1996]. Using this a priori information we estimated site positions, satellite, and receiver clocks, and equivalent zenith propagation delay assuming a horizontally stratified atmosphere at all sites. A free network approach was applied, meaning that no site positions were held fixed during the solution. The receiver clocks were estimated

as white noise processes. To estimate the tropospheric delay, an a priori hydrostatic and wet component were subtracted from the total neutral delay. The residual delay was then estimated as a random walk process with a variance rate of $2.89 \times 10^{-8} \text{ m}^2/\text{s}$, using the Lanyi [1984] mapping function to relate measurements at certain elevation angles to the zenith. The elevation cutoff angle was set to 15° , meaning no observations below an elevation angle of 15° were used in the solutions. The wet delay was then obtained by restoring the total delay from the residual and then subtracting the hydrostatic delay determined from pressure data at the different sites, and the Saastamoinen [1972] formula.

$$\ell_h = (2.2768 \pm 0.0024) \frac{P}{f(\theta, H)} \quad (1)$$

where

$$f(\theta, H) = 1 - 0.00266 \cos(2\theta) - 0.00028H \quad (2)$$

where ℓ_h is the zenith hydrostatic delay in millimeters, P is the pressure in millibars, θ is the site latitude in degrees, and H is station height in kilometers above the ellipsoid. In our case no pressure observations were carried out at the GPS sites, instead we obtained interpolated ground pressure time series derived from the three-dimensional analysis field routinely produced by

Table 2. Values for Coefficients Used in Equation (3)

Coefficient	Value
a_0	5.861
a_1	0.011
a_2	0.054
a_3	0.138

the Swedish Meteorological and Hydrological Institute (SMHI) [Andersson *et al.*, 1986]. Finally, the IWV was calculated from the zenith wet delay using a model especially derived, using radiosonde data from Scandinavia.

$$\frac{\ell_w}{I} = a_0 + a_1 \cdot \theta + a_2 \cdot \sin\left(2\pi \cdot \frac{t_D}{365}\right) + a_3 \cdot \cos\left(2\pi \cdot \frac{t_D}{365}\right) \quad (3)$$

where θ is the site latitude in degrees, and t_D is the decimal day of the year; ℓ_w is the zenith wet delay in millimeters; and I is the IWV in kg/m². The values of the coefficients can be found in Table 2. Figure 2 summarizes the method used to obtain the IWV from GPS data.

3. Independent Techniques for Assessment of Accuracy

3.1. Microwave Radiometer

The water vapor radiometer (WVR) at the Onsala site is a dual frequency microwave radiometer measuring the emission from the sky at 21.0 GHz and 31.4 GHz. From the measured brightness temperatures

the zenith wet delay is calculated according to *Elgered* [1993]. The IWV values are then determined according to (3) above as for the GPS estimates. The WVR was mapping the sky continuously during repeated cycles of 8 min duration for the entire experiment. This procedure resulted in 40 measurements per cycle, i.e., five measurements per minute. The distance between the radiometer and the GPS antenna at Onsala is less than 10 m, and the difference in the heights above mean sea level is less than 1 m.

3.2. Radiosonde

Data from radiosondes were used at four different sites: Sodankylä, Sundsvall, Frösön, and Landvetter. The radiosondes produce vertical profiles of pressure, temperature, and humidity. These profiles were integrated in order to obtain the IWV. Typically two but sometimes four balloon launches per day were available. The radiosondes were of the type Vaisala RS80 which has a reproducibility of < 3% for the humidity sensor.

4. Results

Time series of the IWV were estimated at each GPS site. Table 1 summarizes the GPS data set. The overall variations are typically consistent with the results from nearby sites. A systematic comparison of the results is presented in Table 3, where the mean and standard deviation is shown for each GPS site on a monthly basis. There is an expected systematic increase in the mean IWV as well as in the variability with decreasing latitude. We can also notice the seasonal effect, when the air is getting colder and therefore containing less water vapor as the summer turns into fall.

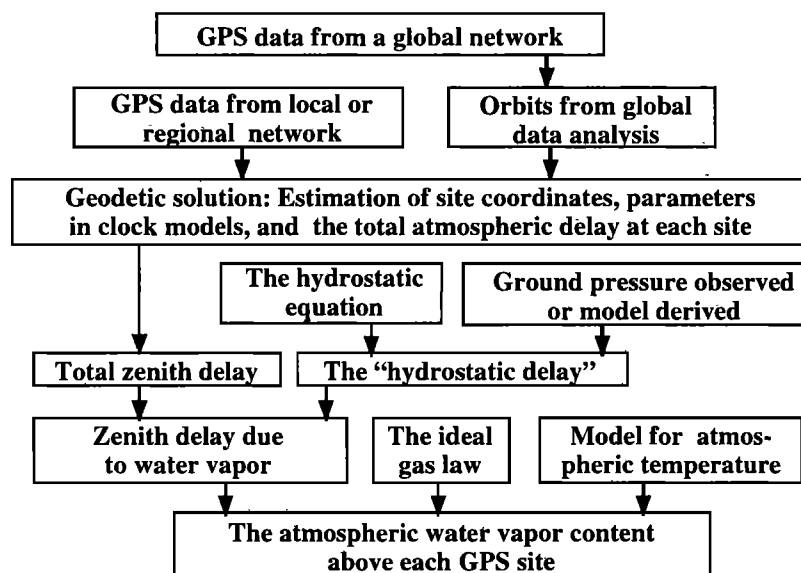


Figure 2. Flow chart over the algorithm used to derive the integrated water vapor from GPS data.

Table 3. Monthly Statistics of the Atmospheric Water Vapor Content Estimated From GPS data

GPS Site	August		September		October	
	Mean kg/m ²	rms kg/m ²	Mean kg/m ²	rms kg/m ²	Mean kg/m ²	rms kg/m ²
KIRU	15.4	3.9	10.4	4.9	8.9	4.4
SODA	16.8	5.1	11.2	4.5	10.4	4.6
ARJE	14.7	4.9	10.3	4.9	9.4	4.6
OVER	18.0	5.4	12.8	4.9	11.9	5.1
OULU	... ^a	...	14.0	3.6	12.6	5.7
SKEL	17.9	5.9	13.4	5.9	12.5	5.7
VILH	15.9	5.3	10.8	5.1	10.8	5.3
UMEA	18.2	5.8	13.6	5.6	13.1	6.2
OSTE	16.9	5.7	12.2	5.1	11.8	5.3
VAAS	... ^a	...	14.7	3.7	13.3	5.9
SUND	19.3	6.5	14.9	6.1	13.6	6.1
SVEG	17.0	5.8	12.8	5.5	11.8	5.3
LEKS	17.3	5.7	13.9	5.5	12.6	5.5
MART	19.4	6.2	15.3	6.2	13.4	6.2
TUOR	18.7	5.5	13.9	6.4	13.8	5.9
METS	17.8	5.4	14.9	6.8	14.2	6.3
KARL	21.6	5.7	17.3	6.1	15.1	6.4
LOVO	20.4	5.4	16.6	5.8	14.5	6.3
VANE	20.0	6.4	18.0	6.2	16.4	6.4
NORR	21.3	5.8	18.9	6.0	15.9	6.6
JONK	20.2	5.2	17.5	5.9	14.8	6.4
VISB	20.3	5.4	17.3	6.7	14.1	6.1
ONSA	18.3	6.3	17.0	5.9	15.3	6.7
OSKA	20.9	5.6	18.9	6.1	15.4	6.7
HASS	20.7	5.9	19.1	5.9	16.4	6.4

^aNo data were acquired at this site for this month.

4.1. Quality Assessment

Figure 3 shows the time series for the three months of August to October 1995 of the GPS-estimated IWV at the Onsala site together with the data measured with the WVR. The top curves show the WVR data offset by 20 kg/m² in order to improve the visual comparison. On the average, the WVR data are 0.9, 1.2, and 1.6 kg/m² above the GPS estimates for the months of August, September, and October, respectively.

Detailed statistics of the WVR-GPS comparison are shown in Figure 4 where day-by-day comparisons between the Onsala WVR measurements and the GPS estimates for the 3 month period are shown. The circles denote the daily mean differences between the data from the two methods, and the bars represent the root-mean-square (rms) scatter of differences about the mean for each day. In order to reduce the WVR measurements to the times of the GPS estimates, we smoothed the WVR data using a Gaussian window with a full width half maximum of 10 min. The time gaps in the figure are due to lack of data from either one of the techniques for the entire or significant part of the day. We can here clearly see a bias between the estimates of about 1 kg/m². Also notable is the very high rms value for the last day in the time series, which we believe could be due to rain, or large drops of liquid water affecting the

WVR measurements. We can also see from the figure that the data from the month of October are somewhat more consistent; the daily mean difference is not varying so much as during the two previous months.

Figure 5 shows the GPS estimated IWV at the Sundsvall Station together with the radiosonde estimates from the same site. The formal error bars on the radiosonde data correspond to 5% of the total value, which is consistent with earlier comparison measurements between radiosondes and microwave radiometry [Elgered, 1993]. We note, as we did in the comparison with the WVR data, that the GPS-estimated IWV seem to be somewhat underestimated. On the average, the radiosonde measurements at Sundsvall are 1.2, 1.5, and 1.2 kg/m² above the the GPS estimates for the months of August, September, and October. The corresponding rms scatter of differences about the mean are 1.9, 1.7, and 1.5 kg/m². Table 4 gives the statistics from the comparisons with radiosonde data from the other sites available. Note that the offset in the Onsala–Landvetter comparison is somewhat larger than for the GPS–WVR comparison. We can, however, expect that the distance Onsala–Landvetter of 38 km contributes some 1–2 kg/m² to the total rms difference. The trend is also consistent between the two comparisons. When we analyze the data summarized in Table 4 in more detail (not shown in the table), we see, as for the GPS–

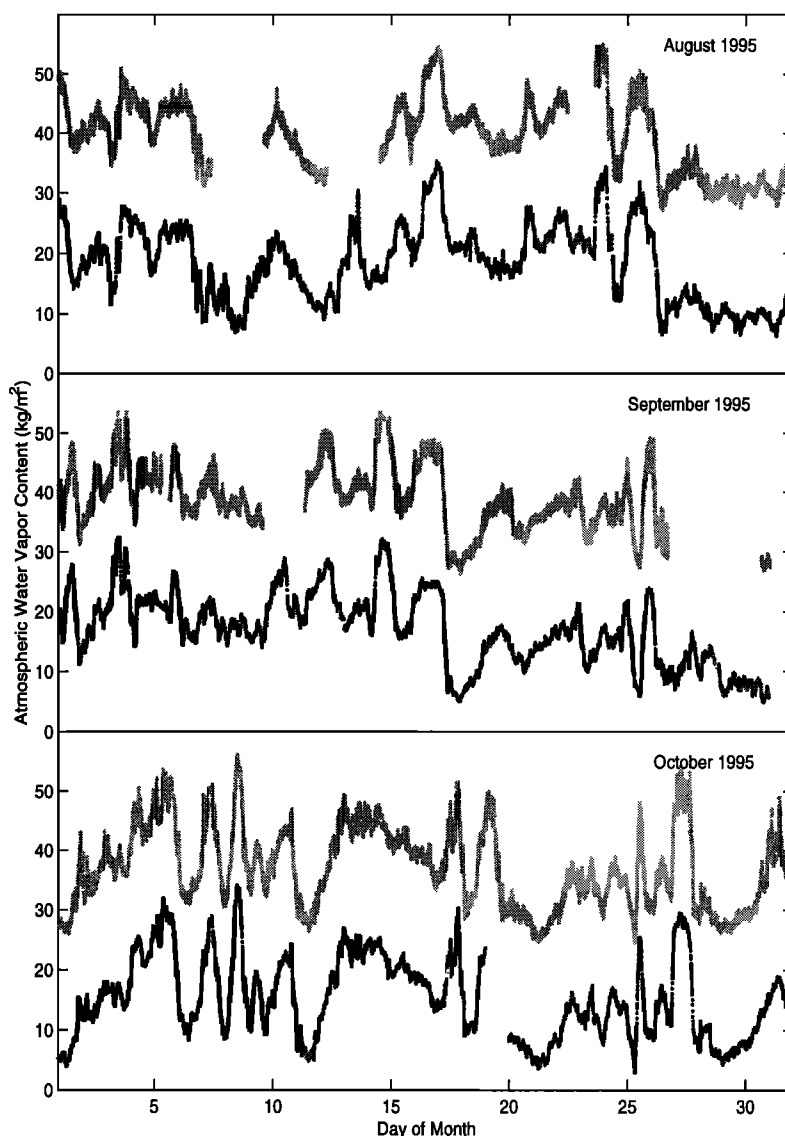


Figure 3. A comparison of the integrated water vapor (IWV) content estimated from GPS and water vapor radiometer data at the Onsala site. Note that the radiometer data have been offset by 20 units.

WVR comparison, that the monthly mean differences between the GPS and the radiosonde data show the largest value in October.

4.2. Spatial Characterization

The spatial structure function of a quantity q is defined as [e.g., *Treuhaft and Lanyi, 1987*]

$$D_q(r) = \langle [q(R+r) - q(R)]^2 \rangle \quad (4)$$

where r is the distance between two sites, and the angle brackets denote the expectation value. *Treuhaft and Lanyi [1987]* presented a spatial structure function for the wet delay quantity. According to their investigation, D will behave as $r^{2/3}$, under the assumption that the spatial structure of the refractive index can be approximated with Kolmogorov turbulence, and r is sufficiently large, ≥ 1 km.

In our calculations we define the spatial structure function for the IWV as

$$D_I(r) = \langle [I(R+r) - I(R)]^2 \rangle \quad (5)$$

However, we use the mean value to approximate the expectation value. By studying radiosonde data alone, we could verify that the power-law dependence on r for the IWV structure functions deviates less than 5% from that of the wet delay structure functions during our study period.

Figure 6 shows D_I as a function of r for the 20 Swedish GPS stations for the month of September 1995. Included in the figure are the corresponding results for the radiosonde stations in the area (Copenhagen, Landvetter, Sundsvall, Luleå, Ekofisk, Jokioinen, and Sodankylä). The straight lines in the figure show the least squares estimates of the power-law dependence

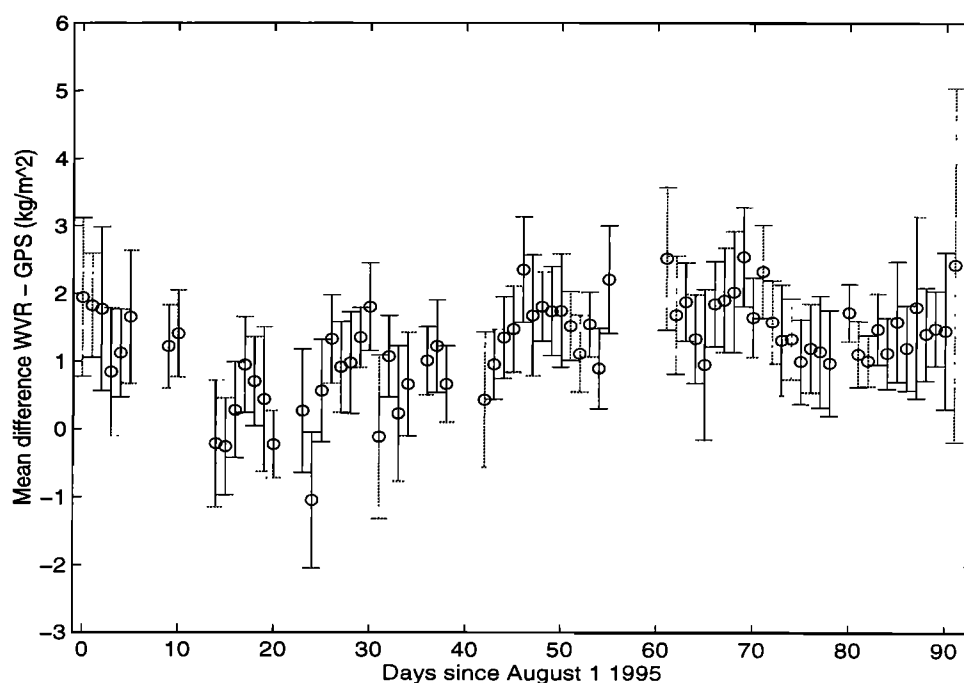


Figure 4. Daily statistics for the comparison between the Onsala WVR measurements and the GPS estimates. The circles denote the daily mean difference between the data from the two methods, and the bars represent the rms scatter of differences about the mean for each day. The abscissa shows the number of days after August 1, 1995.

on r . There is a good agreement between the GPS and the radiosonde structure functions, indicating that the correlations between the site estimates reflect the real atmospheric spatial variations. The estimate of the power-law dependence from the data does not, however, agree with the theoretical value of $2/3$ presented by *Treuhaft and Lanyi* [1987], but are 1.0, and 0.9 for the GPS and radiosonde data, respectively. We thus have a discrepancy between our data and this model, for this time period. We believe the difference in power-law dependence can be due to the fact that the value of $2/3$ is not valid in our climate [*Jarlemark and Elgered*, 1997].

5. Discussion of Errors

The comparisons between the results from the different techniques presented in this paper are affected by different errors. We will give an overview of the error sources specific for each technique.

5.1. GPS-Related Errors

The IWV derived from the GPS measurements can be subject to a variety of errors. The precise ephemerides distributed by the IGS are combinations of orbit solutions from different analysis centers. According to *Beutler et al.* [1996] and *Mireault et al.* [1996] the best four of these contributions show an rms consistency of about 10 cm in satellite coordinates. *Larson et al.* [1991] described the ratio between the error in the baseline length estimate and the orbit error as the ratio between the

baseline length and the distance to the satellite. The effect of orbit errors on the IWV estimates is, however, not thoroughly investigated and should be considered for future studies.

In the data processing the atmosphere was considered to be horizontally stratified, which means that it is azimuthally symmetric. We assume that one equivalent zenith wet delay value determines the wet delay in any direction, given a certain elevation angle. More advanced models, using more parameters to describe the atmosphere, have been proposed as alternatives to this simplified model, for example, by *Davis et al.* [1993] and *Jarlemark and Carlsson* [1996]. Studies have also been carried out by estimating such parameters using the very long baseline interferometry (VLBI) technique [*MacMillan*, 1995]. Using the model presented by *Davis et al.* [1993], we can decompose the zenith wet delay to one azimuth independent term and two gradient components. Figure 7 shows the east-west gradients at Onsala, with the formal one-sigma error bars, estimated from WVR data. We report these gradients using the excess propagation delay. The east-west gradients were typically larger than the north-south gradients. Such systematics were also seen by *Davis et al.* [1993] in an earlier study using WVR data from this site in 1988. From Figure 7 we can see that the size of the gradients is typically less than 2 mm, but periods exist where they can reach up to 10 mm as, for example, during August 15. The effect of this gradient on the IWV estimate can be seen in Figure 8, where the estimate is plotted together

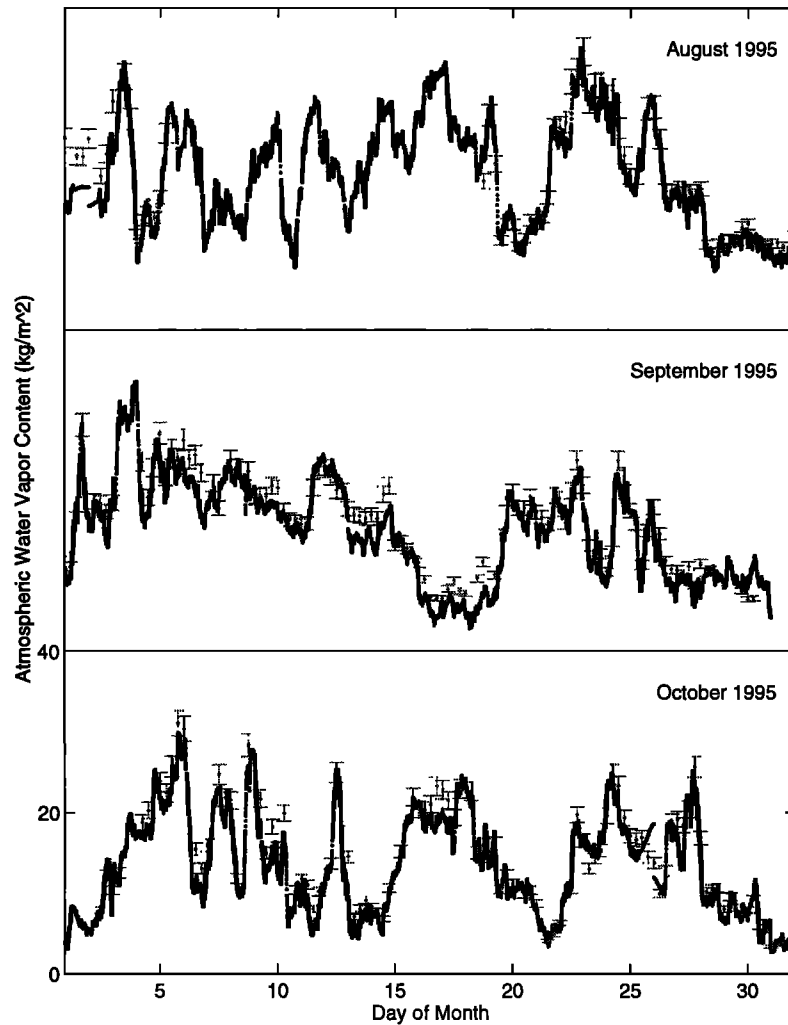


Figure 5. GPS-estimated IWV at the Sundsvall station together with the radiosonde estimates from the same site. The formal error bars on the radiosonde data correspond to 5% of the total value.

with the WVR data. Simulations investigating the effect caused by atmospheric gradients on the wet delay estimates have been carried out by (P. O. J. Jarlemark et al., wet delay variability calculated from radiometric measurements and its role in space geodetic parameter estimation, submitted to *Radio Sci.*, 1997). These simulations indicate that gradients with a size of the order of 1 mm contribute an error of about 0.2 kg/m² to the IWV estimate.

Scattering is the effect of reflections of the signals in the antenna structure or the monument. Using a model described by *Elósegui et al.* [1995], simulations show that we can expect errors in the range of 1–2 kg/m². Because of the different elevation dependence of the atmospheric error and the scattering, the effects could easier be separated by choosing a lower cutoff angle, for example 10° or lower.

Another site specific error source is the radomes used.

Table 4. Distances Between GPS Stations and Radiosonde Sites, and Statistics for the Comparison of IWV Values From Two Methods

GPS / Radiosonde Site	Distance, km	Number of Paired Observations	rms Difference, kg/m ²	Mean Difference, kg/m ²
Sundsvall / Sundsvall	35	268	2.1	1.3
Östersund / Frösön	34	20	1.8	1.4
Sodankylä / Sodankylä	13	159	2.0	0.5
Onsala / Landvetter	38	298	2.6	1.7

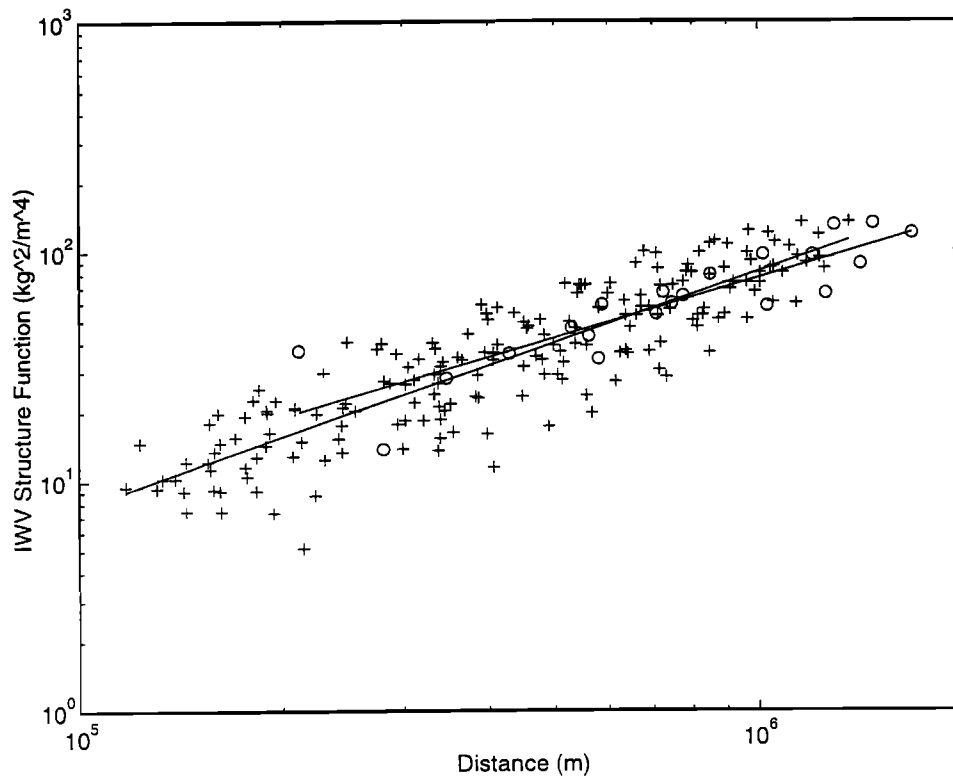


Figure 6. Spatial structure function D_I as a function of r for the 20 Swedish GPS stations for the month September 1995 (plus signs). Included in the figure are also the corresponding results for the radiosonde stations in the area (circles). The straight lines in the figure show the least squares estimates of the power-law dependence on r .

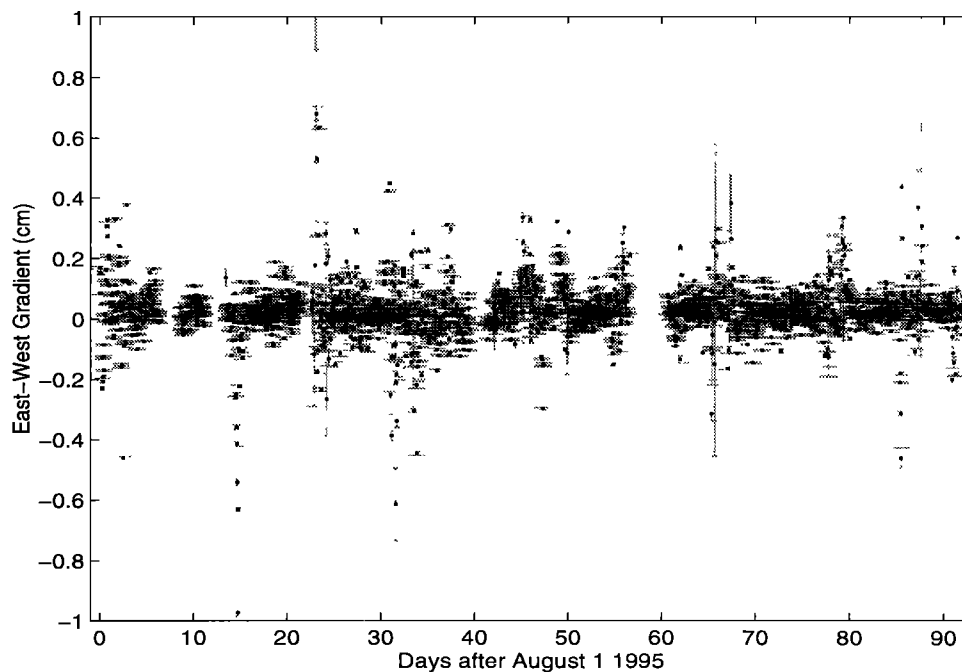


Figure 7. East-west gradients at Onsala with the formal one-sigma error bars estimated from water vapor radiometer (WVR) data.

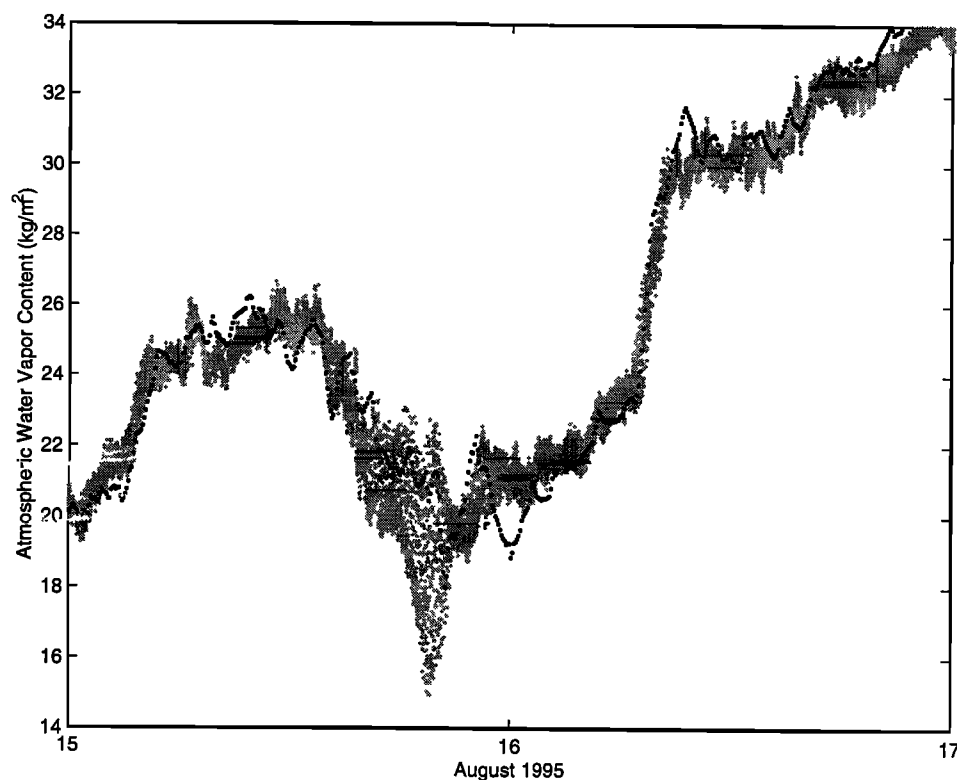


Figure 8. GPS-estimated IWV (black) and corresponding WVR data (gray) for August 15 and 16, 1995. The effect of an atmospheric gradient can be seen in both estimates.

They are used to protect the antennas from rain and snow. This effect is a specific type of scattering. The radome influence and the general scattering effects are therefore difficult to separate unless the measurements can be done both with and without radomes under similar conditions. The 3 month period of this experiment involved three different types of radomes (see Table 1); at the Onsala and the Sodankylä sites the “Delft” type radome was used, whereas a tall conical “white” radome especially designed to prevent snow accumulation was used at all the other Swedish sites. One Finnish site was equipped with the radome delivered with the Ashtech GPS receivers. This one is of a similar design as the “Delft” radome. One method to evaluate the influence of the radomes is to perform so-called elevation cutoff tests. The results from different GPS solutions are compared when using data above different elevation angles. Figure 9 shows an example of an elevation cutoff test, where the difference between the GPS-estimated IWV and the WVR data at the Onsala site for 6 days in the beginning of October 1995 are presented. The two different solutions are carried out with 10° and 15° elevation cutoff angle shown as the black and the light gray curves, respectively. We can see that the two solutions show a similar pattern, the 10° curve however around zero and the 15° around 2 kg/m^2 . Similar results have also been obtained by Bar-Sever and Kroger [1996]. Earlier investigations whose objective was to

study the GPS position estimates have also shown a strong dependence of the estimated vertical position coordinate with respect to the elevation cutoff angle, using data from the Swedish GPS network [Jaldehyag *et al.*, 1996]. Studying the phase data residuals of GPS solutions for sites with “white” radomes also show a clear systematic effect with elevation angle. Because of this all “white” radomes were removed in the summer of 1996. The radome influence was further investigated by using 1 month of GPS data from the Swedish sites during October 1996. During this time period the Onsala site is still equipped with a “Delft” radome, but all other Swedish sites have had the “white” radomes removed. A comparison of the water vapor content with the WVR at the Onsala is shown in Figure 10. We have here a strong indication that the removal of the radomes at the surrounding sites has influenced the water vapor estimates at Onsala. The effects introduced by the radomes will be a subject for further studies.

To remove the delay of the GPS signals introduced by the ionosphere, a linear combination of the two radio signals received is formed. Treating this linear combination as free of ionospheric contribution is a rough approximation, assuming the refractive index depends only on the frequency squared and the electron density. Seeber [1993] describes the refractive index as a power series of $1/f$ where f is the frequency. Ignoring the higher-order terms in the GPS processing results in

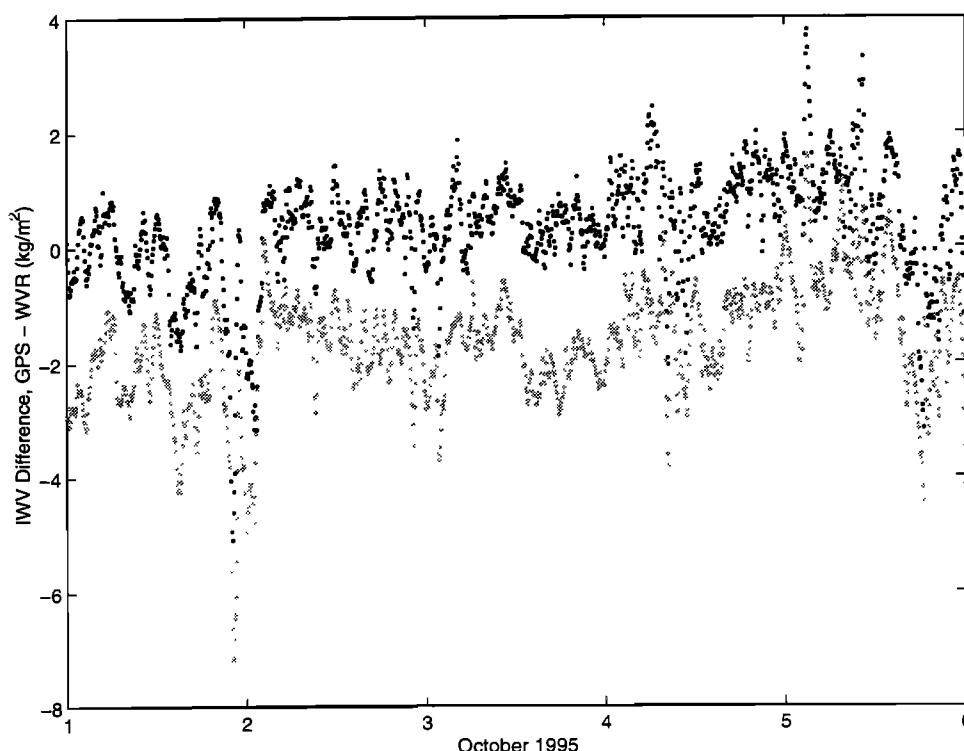


Figure 9. Differences in IWV between GPS estimates and WVR data for 6 days in the beginning of October 1995. The two different solutions are carried out with 10° and 15° elevation cutoff angle shown as the black and the light gray curves, respectively. We can see that the two solutions show a similar pattern, the 10° curve, however, around zero, and the 15° around 2 kg/m^2 .

residual model errors. These errors can be of the order of several centimeters in signal delay during high solar activity [Brunner and Gu, 1991].

The lack of pressure data during the GPS analysis can be the reason for different errors. During the entire GPS processing we assume the GPS receiver to be at a fixed position. This is however not the case since the crust of the Earth is subject to a variety of external and internal influences. These effects have to be modeled, in order not to affect the results. One of these effects is the pressure loading. The vertical position of the GPS receiver changes due to different atmospheric pressure loading the Earth [vanDam and Herring, 1994]. At the Onsala site the daily pressure variations had an extreme value of 15 mbar over 1 day in the data set but was typically below 8 mbar. Assuming the position change to be about 0.2 mm/mbar [Haas and Schuh, 1997], we estimate the error in the final IWV estimate to be typically less than 0.1 kg/m^2 .

A small error in the estimates is introduced in the analysis process by not using the pressure values to calculate the hydrostatic delay before the GPS processing is done. The approach used to withdraw an a priori constant value for the hydrostatic delay will result in a residual delay that is not equal to the wet delay. Treating this quantity as the wet delay considering mapping functions and process parameters is therefore somewhat

misleading. However, simulations show that using the pressure readings in the post processing, the error in the final IWV estimates due to this technique is $\leq 0.2 \text{ kg/m}^2$.

The choice of process parameter for the estimation of the water vapor in the GPS processing is for this processing both time and site independent. The actual fluctuations of the water vapor in the atmosphere are, however, dependent on the site position as well as the time of the year. Simulations by (P. O. J. Jarlemark et al., wet delay variability calculated from radiometric measurements and its role in space geodetic parameter estimation, submitted to Radio Sci., 1997) show that rms errors in the IWV estimate can be of the order of 0.5 kg/m^2 when we have a very high atmospheric variability but are more likely to be around 0.3 kg/m^2 during normal circumstances. Similar results were obtained by Bar-Sever and Kroger [1996], comparing GPS-estimated IWV with WVR data.

The encryption of one of the satellite-transmitted codes, Anti-Spoofing (AS), was on during the entire time period of the data set. We do not, however, consider this effect to significantly affect the quality of our estimates.

5.2. WVR-Related Errors

The WVR data are as opposed to the GPS estimates severely affected by rain, or large drops ($> 0.5 \text{ mm}$)

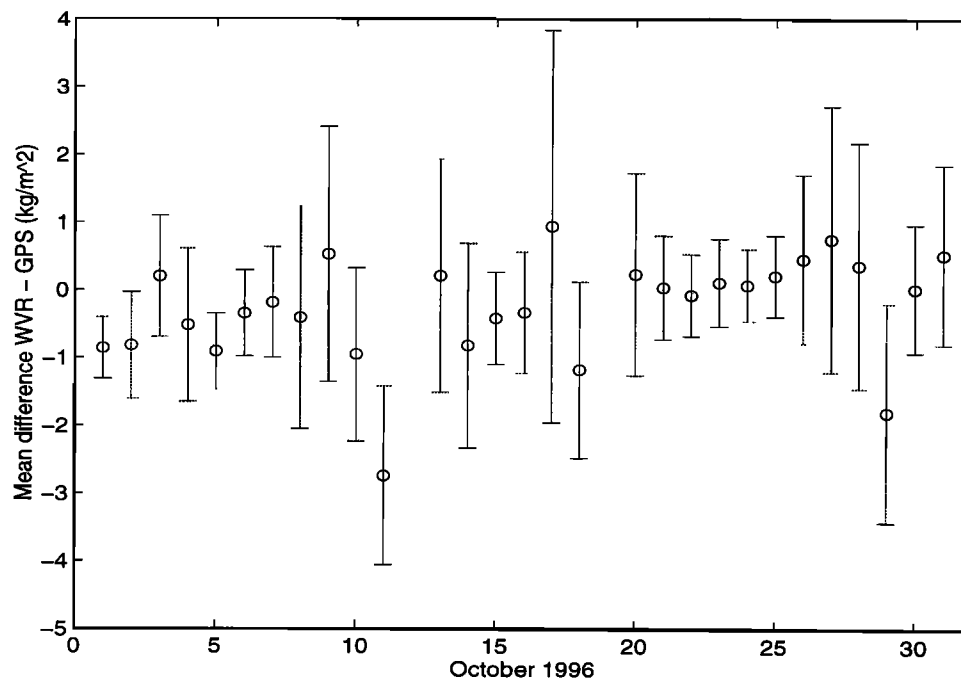


Figure 10. Daily statistics from a GPS, WVR comparison of the same type as presented in Figure 4. This data set is from October 1996, when the “white” radomes were removed from the Swedish GPS sites. Also in this case an elevation cutoff angle of 15° was used. The mean difference (WVR - GPS) for this month is -0.4 kg/m^2 .

of liquid water in the atmosphere. For the comparisons presented here, data from the WVR have been removed when the liquid content exceeds 0.7 kg/m^2 . This amount of liquid water corresponds to about 1 mm in delay [Elgered, 1993] and is somewhat higher than that used by, for example, Coster *et al.* [1996]. However, we can still observe periods in the WVR data, for example, the last day in October (see Figure 3), when fluctuations in the IWV are very high. These fluctuations sometimes occur during rain periods, see Rubel [1996] for a comprehensive presentation of the precipitation in the Baltic area during this experiment. However, we cannot exclude that the variations are real IWV fluctuations, which can be expected to be high also before and after periods of convective rainfall. A lower value for this threshold would probably improve the agreement between the WVR data and the GPS estimates, since more periods with large short-time variations would be removed from the comparison. The uncertainty in the WVR data are assumed to be of the order of 5% of the total value. For this data set this means uncertainties of up to 2 kg/m^2 but more typically around 1 kg/m^2 .

5.3. Other Errors

Uncertainty in the pressure values used to calculate the hydrostatic delay will, of course, also affect the estimated IWV. An error of 1 mbar in the pressure at a site will result in an error of 0.3 kg/m^2 in the IWV estimate. Comparisons of the interpolated pressure time series ob-

tained for the Onsala site with the independent pressure readings at the site show an rms difference of 0.45 mbar, 0.38 mbar, and 0.38 mbar, respectively, for the months of August, September, and October. Considering those results we assume the errors due to erroneous pressure values to be less than 0.2 kg/m^2 .

The uncertainty of 0.0024 mm/mbar in (1) when calculating the hydrostatic delay corresponds to about 0.35 kg/m^2 in the IWV. The error introduced by the assumption of hydrostatic equilibrium is typically around 0.01% of the hydrostatic delay corresponding to 0.03 kg/m^2 but may vary and can be as high as 1 kg/m^2 under extreme conditions [Hauser, 1989].

The factor ℓ_w/I from (3) has an rms uncertainty of about 1%. This gives an rms uncertainty of around 0.4 kg/m^2 in the worst case for the final IWV. Using radiosonde data, for which we can calculate both IWV and zenith wet delay values, we can compare these two quantities to get an indication of the size of these errors during the period of study. This effect was largest for the data from the Landvetter site, among the four sites used in the comparisons. There we can see an rms difference of 0.25 kg/m^2 between the direct IWV measurements and the IWV values determined from the zenith wet delay. We should also note that this error is usually correlated with other GPS-estimate errors. At, for example, the Sundsvall site the covariance was $0.2 \text{ kg}^2/\text{m}^4$. The error is, however, comparatively small, and a comparison between the GPS estimates and the

Table 5. Possible Errors Affecting the WVR and GPS Estimates of the IWV

Error	Typical kg/m ²	Maximum kg/m ²
Orbital	?	?
Gradients	0.2	2
Scattering	...	1
Radomes	...	2
Ionosphere	?	?
Pressure loading	0.1	0.4
Mapping function	?	?
Process parameters	0.3	0.5
Pressure correction	0.1	0.2
Hydrostatic equilibrium	0.03	1
WVR uncertainty	0.9	2
"Delay-to-IWV" alg. ^a	0.25	1
Total root-sum-square ^b	≈1.0	
Total maximum		≈9

The values given for the different errors are the approximate numbers for maximum and typical errors.

^aThese errors are not included in the summation because both the WVR and the GPS results were obtained with the same algorithm in our comparisons.

^bAssuming that the "typical" errors are uncorrelated 1σ Gaussian.

IWV calculated from the radiosonde zenith wet delay values using (3), instead of using the radiosonde IWV values directly, shows rms differences deviating ≤ 0.1 kg/m² from the statistics shown in Table 4.

Table 5 summarizes the possible errors affecting the WVR and GPS-estimates of the IWV. Errors due to orbits, ionosphere, and mapping functions are believed not to dominate, but more detailed studies may be important if an accuracy better than 1 kg/m² is desired.

6. Conclusions and Discussion

We have shown that it is possible to use a permanent network of GPS receivers to estimate the amount of atmospheric water vapor on a mesoscale level. Comparing the GPS estimated IWV with values derived from the WVR shows an agreement of 1–2 kg/m². Much of the rms difference between the estimates can be derived from the slowly varying bias between the two data sets. It should be possible to reduce this bias since the level of the GPS estimates are significantly influenced by the choice of, for example, the radomes used in combination with the elevation cutoff angle. Taking this into consideration, we can estimate the quality of our GPS estimates by looking at the rms scatter of the observed difference around their mean value. This is the same quantity as the total rms difference if the bias specific for each day was removed, which can be regarded as a realistic goal when trying to improve the accuracy. We note, however that this number includes errors both from the GPS and from the WVR technique. The mean of the daily values for this measure is 0.8 kg/m². Another interesting application of the GPS-derived IWV is

in combination with the technique using GPS receivers in low Earth orbits, so-called occultation [e.g., *Hajj et al.*, 1996; *Feng et al.*, 1996]. Combining these two techniques would improve, we believe, the possibilities of obtaining information on the vertical profiles of the atmospheric water vapor and temperature.

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